

Functional & Non-Functional Requirements

Target Audience

Our target audience consists of commuters and drivers in Singapore who are seeking reliable and convenient ways to plan their journeys. They value real-time information on carparks, traffic conditions, bus arrivals, and taxi availability, all accessible through a single platform. The system also caters to users who prefer simple, intuitive tools that save time and provide cost-effective travel recommendations.

Functional Requirements

1. Website must provide account creation for first time users using verifiable information
 - 1.1. Account creation must support two flows:
 - 1.1.1. OAuth-based sign-up using Google authentication.
 - 1.1.2. Email, Password and OTP sign-up
 - 1.2. Information entered should be in the correct format
 - 1.2.1. OTP must match the 6-digit code sent to the user's email, if any.
 - 1.2.2. Email must be in the standard format.
 - 1.2.3. Password must be at least 8 characters long,
 - 1.2.3.1. Password must include at least one lower case
 - 1.2.3.2. Password must include at least one upper case
 - 1.2.3.3. Password must include either a special character or number.
 - 1.3. Website must prohibit the creation of account if any of the information does not meet the requirements.
 - 1.4. Verification email must be sent to the registered email address.
2. Website must allow registered users to log in to their account
 - 2.1. Website must provide error message when incorrect login credentials are entered
 - 2.2. Website must allow registered users to reset their password if they forget it
 - 2.2.1. Website must send an email to registered users containing a link for them to reset their password.
 - 2.3. Website must allow users to log out at any time after logging in.
3. Website must request permission to access the user's location.
 - 3.1. If location is granted, zoom in to display the user's current location and their desired destination along with the relevant information on the map view.
 - 3.1.1. Website should display nearby carparks (if any) with icons.
 - 3.1.2. Website should display nearby and on-route roadworks (if any) with icons.
 - 3.1.3. Website should display the taxis around the area (if any) with icons.

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- 3.1.4. Website should display the nearby and on-route bus stops (if any) with icons.
- 3.1.5. Users must be able to click on the carpark icon to view the following details
 - 3.1.5.1. Name of the carpark
 - 3.1.5.2. Number of available lots
 - 3.1.5.3. Type of lots (e.g. multi-storey/surface/sheltered/basement etc)
 - 3.1.5.4. Users must be allowed to click refresh to get the latest number of available lots.
- 3.1.6. Users must be able click on the bus stop icon to view the following details
 - 3.1.6.1. Bus stop name and code
 - 3.1.6.2. Bus arrival timings
 - 3.1.6.3. Bus diversions (if any)
 - 3.1.6.4. Users must be allowed to click refresh to get the latest bus arrival timings.
- 3.1.7. Users must be able to click on traffic alert icons to view the following details
 - 3.1.7.1. Type of alert (lane closure/roadwork/accident etc)
 - 3.1.7.2. Affected roads
 - 3.1.7.3. Alert descriptions
 - 3.1.7.4. Users must be allowed to click refresh to get the latest alerts or if traffic alert has been cleared.
- 3.1.8. Clicking on taxi icons does not do anything. They are just there for visual density.
- 3.2. If location is not granted, the website must show a zoomed-out map of Singapore.
 - 3.2.1. Website must show the general traffic, carpark, taxi hotspots and bus stop data if the user zooms in to a specific part of the map using icons.
 - 3.2.2. Clicking on the carpark, bus stop and roadwork icons should show the same information as in 4.1.5, 4.1.6 and 4.1.7 respectively.
- 4. Website must pull information directly from the LTA API to ensure reliability and accuracy of data.
 - 4.1. Website should update the information every minute or whenever the respective icons are clicked to ensure constant accuracy of data.
 - 4.1.1. Information includes bus arrival timings, taxi availability locations, carpark availability and traffic alerts.
- 5. Website must provide a search functionality.
 - 5.1. Website must allow users to search for carparks by location or postal code.
 - 5.2. Website must allow users to search for bus stops by name or code.
 - 5.3. Website must allow users to view a list of traffic alerts.
 - 5.4. Website must show the appropriate message if no results are found.

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6. Website must provide AI-powered travel type recommendations.
 - 6.1. Users must be able to enter an end location into a search bar
 - 6.2. Based on current conditions, derive and suggest optimal travel mode using developed AI model.
 - 6.3. Only 3 options are available:
 - 6.3.1. Car – if roads are clear and parking is available
 - 6.3.2. Taxi – if availability is high and roads are congested
 - 6.3.3. Bus – if taxi is too expensive and the place is easily accessible using bus.
 - 6.4. Recommendations must consider:
 - 6.4.1. Real time traffic conditions
 - 6.4.2. Estimated costs for all modes
 - 6.4.3. Estimated travel time for all modes.
7. Website must have a user profile page
 - 7.1. In the profile page, the website must display the user's email and phone number.
 - 7.2. Website must allow users to change their password.
 - 7.3. Website must allow users to log out.
8. Website must allow the user to choose between different map views from the Navigation Bar.
 - 8.1. If user clicks any of the buttons, they are taken to a new screen consisting of a list menu and general map.
 - 8.2. User can choose Bus view.
 - 8.2.1. User can input a destination and view relevant bus stops on the selected path.
 - 8.2.2. User can also view total price, and estimated time of arrival based on the AI's analysis.
 - 8.3. User can choose Taxi view;
 - 8.3.1. User can input a destination and view nearby taxis, current prices, and estimated time of arrival on the AI's analysis.
 - 8.4. Users can choose Drive view.
 - 8.4.1. User can input a destination and view relevant carparks nearby, estimated time of arrival, ERP charges, and relevant traffic alerts based on the AI's analysis.
 - 8.5. All textual and numeric information is displayed as a list beside a general map view.
9. Website must allow the user to click "Start Now" on the landing page to bring them to the general map view.

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9.1. Users must be able to input a destination and view travel options (Bus, Taxi, Drive) with corresponding details as in Section 8.

9.2. All textual and numeric information is displayed as a list beside a general map view.

Non-functional Requirements

1. Performance

- NFR1.1: The website must load within 3 seconds on a standard broadband connection.
- NFR1.2: The system must be able to return search results and LTA API data within 5 seconds.

2. Usability

- NFR2.1: The interface must be intuitive and easy to use, with clear icons and labels.
- NFR2.2: Error messages (e.g., invalid login, no search results) must be clear, descriptive, and actionable.
- NFR2.3: The tutorial (for first-time users) must be short, concise, and skippable, ensuring users can start quickly without frustration.

3. Accessibility

- NFR3.1: The website must follow basic WCAG 2.1 standards for accessibility.
- NFR3.2: Users must be able to navigate important features using only the keyboard.

4. Reliability

- NFR4.1: The system must maintain 99.9% uptime to ensure reliability.
- NFR4.2: The data presented in the system must be updated regularly to reflect the latest available information.

5. Scalability

- NFR5.1: The system must support up to 5,000 users at the same time.
- NFR5.2: The system must be scalable to accommodate additional data sources and support future growth in users and features.

6. Security

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- NFR6.1: User data must be encrypted in storage and during transmission.
- NFR6.2: Password reset, and account verification must be done securely through email links.

7. Compatibility

- NFR7.1: The website must work on the latest versions of Chrome, Firefox, Safari, and Edge.
- NFR7.2: The website must be responsive and adjust to mobile, tablet, and desktop devices.

8. Maintainability

- NFR8.1: The system's code must be modular and documented for easy updates.
- NFR8.2: Bug fixes and updates must be deployable with minimal downtime.